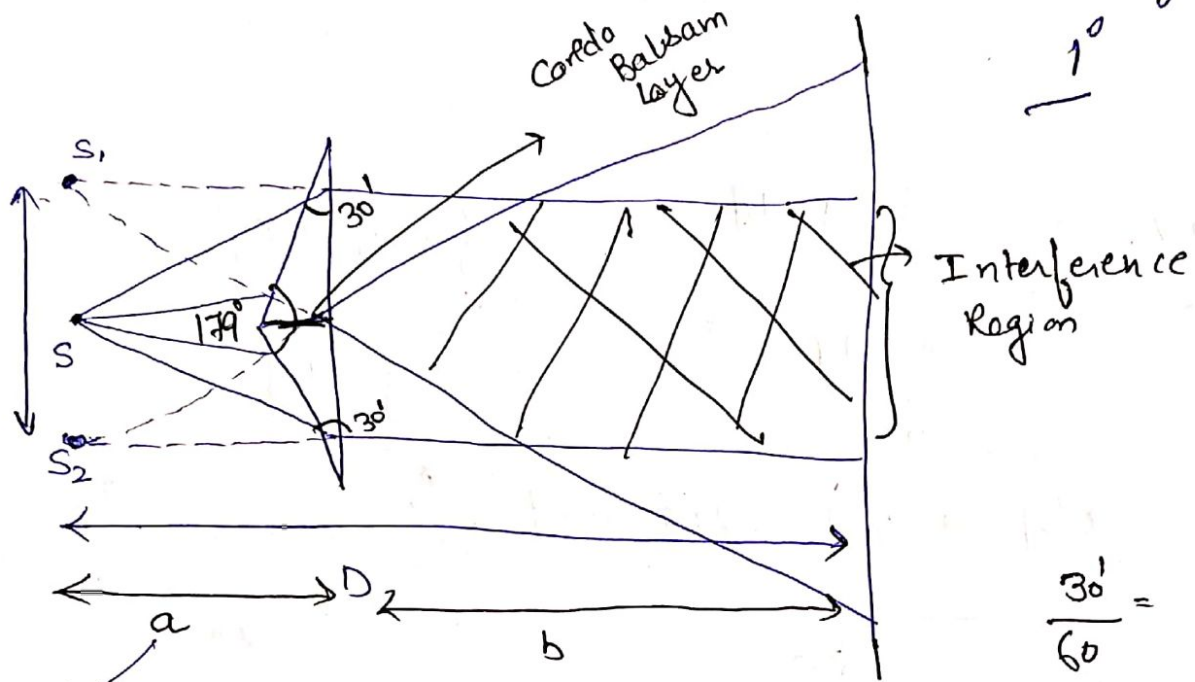


Fresnel's Biprism:-

Combination of two base to base connected prism

one obtuse angle 179°

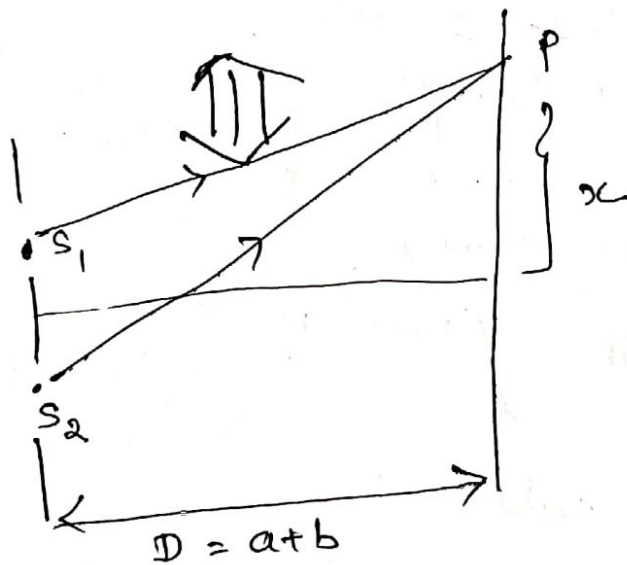


distance b/w sources & biprism

$$\Rightarrow D = a + b$$

$$\frac{30'}{60} =$$

$$\frac{30''}{3600} =$$



$$\Rightarrow \beta = \frac{\lambda D}{2d}$$

$$\beta = \frac{\lambda(a+b)}{2d}$$

Application of wavelength of Sodium light using biprism.

$$\lambda = \frac{\beta \cdot 2d}{(a+b)}$$

from eye piece
or $\beta = \frac{\text{Distance moved}}{\text{Number of fringes moved across the cross wire}}$

Distance b/w two virtual sources i.e. $2d$:-

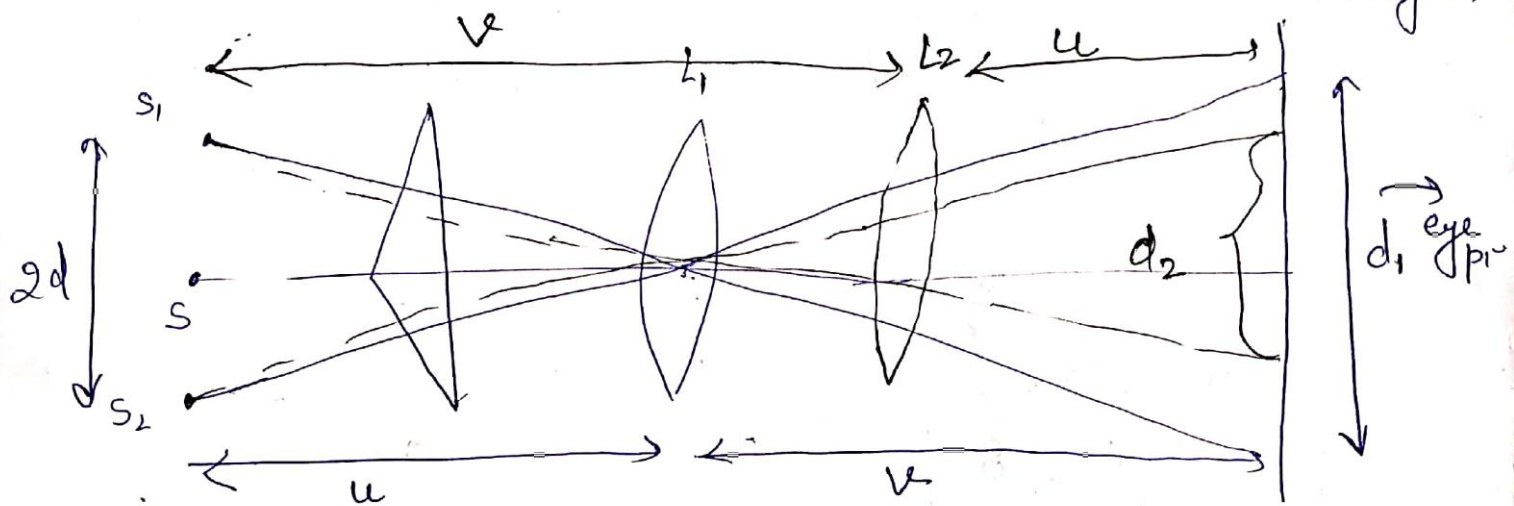
1. Magnification method or Displacement Method :-

In this method, a convex lens L_1 having focal length less than one-fourth of the distance b/w slit and eye piece

$$f < \frac{D}{4}$$

$$\Rightarrow \boxed{D > 4f}$$

is placed in b/w biprism and screen. It must be adjusted like that we can see the sources i.e. S_1 & S_2 images.



$$\text{magnification} = \frac{v}{u} = \frac{d_1}{2d} \quad \text{(Size of image)} \quad \text{--- (1)}$$

$$\text{(Size of object)}$$

Now for the second position of lens means at L_2 again we get the image of sources now

$$\text{magnification} = \frac{u}{v} = \frac{d_2}{2d} \quad \text{--- (2)}$$

From (1) & (2)

$$\frac{v}{u} \times \frac{u}{v} = \frac{d_1}{2d} \times \frac{d_2}{2d}$$

$$\Rightarrow \boxed{2d = \sqrt{d_1 d_2}}$$

2. Deviation method

As we know Refracting angle of prism $\alpha = 30^\circ$ and when light passes through prism, it will deviate by angle δ

and with this deviation we can find out

$$\mu = \frac{c}{v}$$

$$\Rightarrow c = \underline{\mu v}$$

μ by prism formula

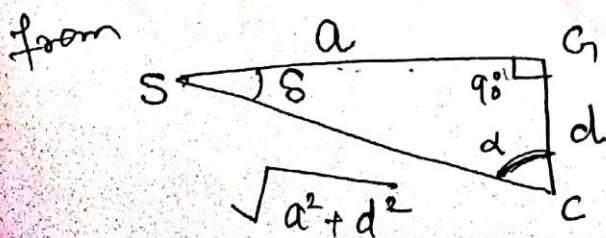
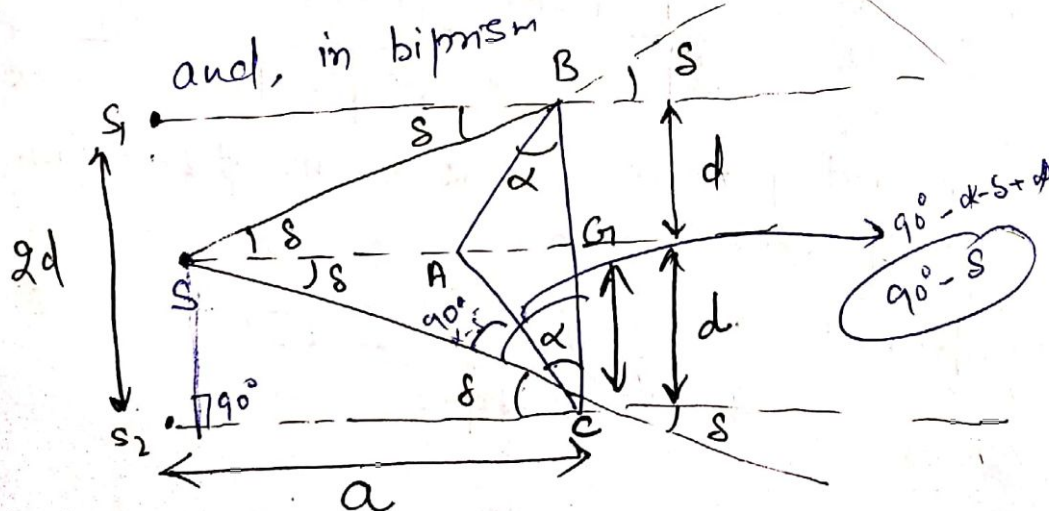
$$\mu = \frac{\sin\left(\frac{\alpha + \delta}{2}\right)}{\sin\left(\frac{\alpha}{2}\right)}$$

both are small
so for small angle
 $\sin \theta \approx \theta$

$$\approx \frac{\frac{\alpha + \delta}{2}}{\frac{\alpha}{2}}$$

$$\mu \alpha = \alpha + \delta$$

$$(\mu - 1)\alpha = \delta$$



distance b/w S_1

$$\left(\frac{d}{a}\right) = \tan \delta$$

$$d = a \delta$$

Similarly distance b/w SS_2

$$\frac{d}{a} = \tan \theta$$

$\therefore \theta$ is small

$$\Rightarrow d = a \theta$$

Now total distance $SS_1 + SS_2 = d + d = a\theta + a\theta$
 $\Rightarrow 2d = 2a\theta$

$$\# \quad 2d = 2a(4-1)\alpha$$

Effects $\rightarrow$

① with increasing angle of biprism:-

$$\text{if } \alpha \uparrow \Rightarrow 2d \uparrow$$

and $\beta = \frac{\lambda D}{2d} \downarrow \Rightarrow \beta$ decrease
 $\Rightarrow$ separation b/w fringes dec. and overlapping is there.

② with increase in slit width

$2d \uparrow \rightarrow$ then distance b/w sources inc. violet
 the condition of sustained interference

and consider, equivalent to pairs of narrow slits and contrast will be poor b/w maxima & minima.

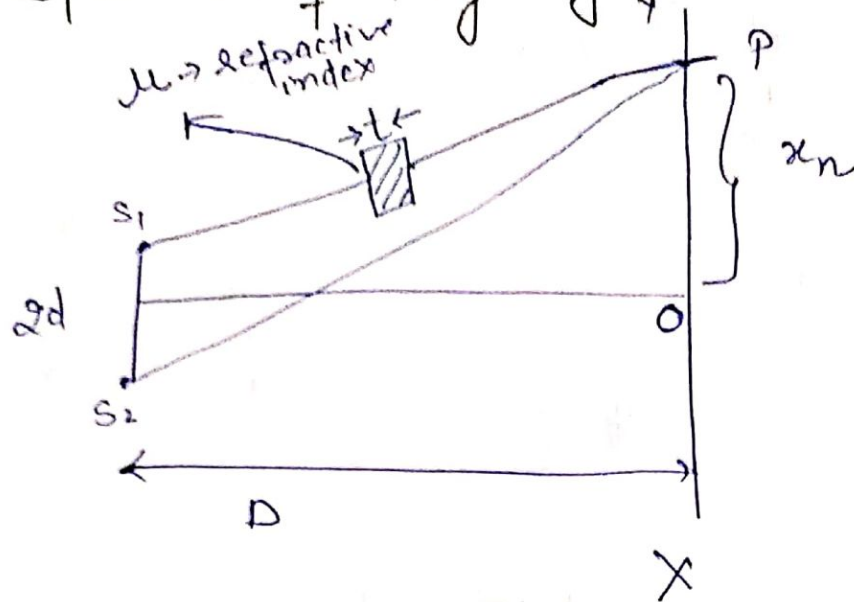
③ with use of white light:-

$$\lambda = 4000 \text{ \AA} \rightarrow 7000 \text{ \AA}, \Delta = n\lambda \cdot \text{maxima} \Rightarrow \Delta = 2n\lambda \cdot \text{minima}$$

Then interference pattern consists of a central white fringe having a few coloured fringes on both sides with a general illumination.

④ To locate the fringe of zero-order:- if use monochromatic, all bright fringes are of same colour, so, it is n't possible to locate the zero-order fringe or central maxima then white light must be used so that we can fix cross wire on central white fringe then we can further calculate the fringes

Displacement of fringes by Introduction of a thin film



time taken by light from S_1 to P

$$\therefore \odot = \frac{d}{t}$$

$$t = \frac{d}{C}$$

$$T_1 = \frac{S_1P - t}{C} + \frac{t}{v}$$

$$\text{or } \mu = \frac{C}{v} \Rightarrow v = \frac{C}{\mu}$$

$$T_1 = \frac{S_1P - t}{C} + \frac{\mu t}{C}$$

$$C \times T_1 = \text{Path travelled by light from } S_1 \text{ to P}$$

$$S_1P + (\mu - 1)t$$

then

$$T_2 \rightarrow \text{from } S_2 \text{ to P}$$

$$T_2 = \frac{S_2P}{C} \Rightarrow S_2P = C \times T_2$$

Path difference

$$(S_2P - S_1P) - (\mu - 1)t$$

$$\text{effective path diff} = \frac{2x_n d}{D} - (\mu - 1)t$$

$$\Rightarrow \frac{2d}{D} x_n - (\mu - 1)t = n\lambda$$

$$\Rightarrow x_n = \frac{D}{2d} [n\lambda + (\mu - 1)t]$$

↓
Position of n^{th} maximum Φ
in presence of film
or sheet.

If sheet is absent, then $t = 0$

$$x_n = \frac{n\lambda D}{2d}$$

$\Rightarrow$ the displacement in n^{th} bright fringe is
or shifting

$$\Rightarrow \Phi(x_n)_{\text{with film}} - (x_n)_{\text{without film}} = x_0$$

$$x_0 = \frac{D}{2d} (\mu - 1)t$$

As $\mu > 1 \Rightarrow x_0 = +ve \Rightarrow$ Shifting upward.

As it is independent of n , so shifting is same for entire fringe system (means shifting is also exists in case of dark fringes).

→ Applications: →

① Determination of thickness of micro plate.

② Determination of fringe width.

$$x_0 = \frac{\beta}{\lambda} (\mu - 1)t$$

$$\therefore \beta = \frac{\lambda D}{2d}$$

$$\Rightarrow \frac{\beta}{\lambda} = \frac{D}{2d}$$